N 1

$$U_n(x)U_m(x) = \frac{\sin((n+1)\varphi)\sin((m+1)\varphi)}{\sin^2\varphi} =$$
$$= \frac{\cos((n-m)\varphi) - \cos((n+m+2)\varphi)}{2\sin^2\varphi}$$
$$S = \sum_{j=0}^{n} \cos(r-2j)\varphi; \quad 2\sin\varphi\cos\alpha = \sin(\alpha+\varphi) - \sin(\alpha-\varphi)$$
$$\Rightarrow 2S\sin\varphi = \sin((r+1)\varphi) - \sin((-r-1)\varphi) = 2\sin((r+1)\varphi)$$
$$\Rightarrow S = \frac{\sin((n+1)\varphi)}{\sin\varphi}$$
$$\sum_{k=0}^{m} U_{n+m-2k}(\cos\varphi) = \frac{1}{\sin\varphi}\sum_{k=0}^{m}\sin((n+m-2k+1)\varphi)$$
$$\sum_{k=0}^{m}\sin((n+m-2k+1)\varphi) = |2\sin\varphi\sin\alpha = \cos(\alpha-\varphi) - \cos(\alpha+\varphi)|$$
$$= \frac{1}{2\sin\varphi}(\cos((n-m)\varphi) - \cos((n+m+2)\varphi)) =$$
$$= \frac{\sin((m+1)\varphi)\sin((n+1)\varphi)}{\sin\varphi}$$
$$\Rightarrow \sum_{k=0}^{m} U_{n+m-2k}(\cos\varphi) = \frac{\sin((m+1)\varphi)\sin((n+1)\varphi)}{\sin^2\varphi} = U_m(x)U_n(x)$$

N 2

$\forall x, y \in H : \langle Tx, y\rangle = \langle x, T^*y\rangle$, где T^* – сопряженный оператор

а) $H = \ell^2, \quad T(x_1, \dots) = (0, x_1, x_2, \dots)$

$$\langle Tx, y\rangle = \langle (0, x_1, x_2, \dots), (y_1, y_2, \dots)\rangle = \sum_{n=1}^{\infty} x_n\overline{y_{n+1}} =$$
$$= \langle x, (y_2, y_3, \dots)\rangle \Rightarrow T^*(y_1, y_2, \dots) = (y_2, y_3, \dots)$$

б) $H = L^2([0,1], dx), \quad T(f)(x) = \int_0^x f(t)dt$

$$f, g \in H \Rightarrow \langle Tf, g\rangle = \int_0^1 \left(\int_0^x f(t)dt\right)\overline{g(x)}dx = \int_0^1 f(t)\left(\int_t^1 \overline{g(x)}dx\right)$$
$$dt = \int_0^1 f(t)\left(\overline{\int_t^1 g(x)dx}\right)dt = \left\langle f, t \mapsto \int_t^1 g(x)dx\right\rangle \Rightarrow$$
$$\Rightarrow (T^*g)(x) = \int_x^1 g(t)dt$$